How the two-ρ terms give JIMWLK and BFKL

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1 The result in short

Notation, as in your notes:

$$\mathcal{N}_0 \equiv \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}, \qquad L_\Lambda \equiv \log\frac{\vee}{\Lambda}, \qquad L_k \equiv \log\frac{\vee}{k^+}, \qquad \mathcal{N} = \mathcal{N}_0 L_\Lambda. \tag{1.1}$$

- $\mathcal{J}$ is JIMWLK⊗LO, the JIMWLK evolution of the LO cross section per unit rapidity (Sec. 3).
- $\mathcal{B}$ is BFKL⊗LO, the BFKL evolution of the LO cross section per unit rapidity (Sec. 6).

The two-ρ terms at leading logarithm are

$$\boxed{\left.\frac{d^3N}{d^3k}\right|^{2\rho}_{\rm LL} = \mathcal{N}_0\Big[L_\Lambda\,\mathcal{J} + L_k\,(\mathcal{B}-\mathcal{J})\Big] = \mathcal{N}_0\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}\Big].} \tag{1.2}$$

This note shows the two halves explicitly.

1. JIMWLK: the coefficient of $\log(\vee/\Lambda)$ (Part I). It is the sum of three sets of two-ρ terms:

(a) the two-ρ terms of the three-ρ equation (`TwoRho_from_final.tex`, Eq. (1), 335 terms);

(b) the genuine two-ρ terms of `Evolution_two_rho.tex`, Eq. (2.1) (75 terms);

(c) the two-ρ lists of `4rho.tex`, Eq. (2.2) (30 terms).

Added up kernel structure by kernel structure, they give exactly JIMWLK⊗LO, Eq. (3.2). This holds in all 13 kernel structures and in all 50 independent colour structures with a nonzero entry. The coefficients are exact polynomials in N_c.

2. BFKL: the coefficient of $\log\vee$ (Part II). Both of your logarithms contain $\log\vee$. So the coefficient of $\log\vee$ is

$$P = [\text{coefficient of } \log(\vee/\Lambda)] + [\text{coefficient of } \log(\vee/k^+)].$$

Row by row, P is the integrand of the p^+ integral at its $\vee$ end, where the unobserved gluon is harder than the measured one. The P two-ρ terms of all rows are listed in Eq. (5.1) (300 terms). Their sum is exactly BFKL⊗LO, Eq. (6.2), in all 9 kernel structures and 49 independent colour structures with a nonzero entry.

Why the split is like this (shown in `TwoRho_all.pdf`, Sec. 3, without taking any limit).

- Λ enters a row only through the limits of its p^+ integral where a gluon carries momentum Λ. That gluon is softer than the measured one: it evolves the target (JIMWLK).
- $\vee$ enters only through the upper limit, where the unobserved gluon is harder than the measured one: it evolves the projectile (BFKL).
- Of your two logarithms only $\log(\vee/\Lambda)$ contains Λ. So the coefficient of $\log(\vee/\Lambda)$ is JIMWLK. BFKL is the coefficient of $\log\vee$, which sits in both of your logarithms.

Variables.

- The measured gluon is at w (amplitude) and w' (c.c.); the unobserved gluon is at z; the two charges are at x and x'.
- x is tied to w and x' to w' by their emission kernels. If one charge emits at both w and w', it is x'.
- $\mathcal{K}(X) = X/X^2$. The phase $e^{-ik(w'-w)}$ and the integrals over x, x', z are understood.
- The two charges are a symmetric product, $\rho^a(p)\rho^b(q) \equiv \frac{1}{2}\{\rho^a(p), \rho^b(q)\}$.
- The colour algebra uses $U^{bc} = U^{cb}$.

Part I
JIMWLK: the coefficient of $\log(\vee/\Lambda)$

2 The three sets of two-ρ terms

2.1 (a) From the three-ρ equation

These are the 335 terms of `TwoRho_from_final.tex`, Eq. (1). They are obtained by reordering every black and blue line of `ThreeRho_final.tex`. They are used here exactly as written there.

2.2 (b) The genuine two-ρ terms

These are your terms of `Evolution_two_rho.tex`, in the same notation.

- *Group I, Row II.* The bracket $-\frac{2}{\epsilon} - 2\gamma - \log\frac{(x-w)^2\mu^2}{4}$ is $\frac{1}{\pi}\int_z[K(x,w,z) - K(w,w,z)]$ (fix F6). Only its pole and constant change: $\frac{1}{\pi}\int_z K(a,b,z) = \frac{1}{\bar\epsilon} - \log(4\pi^2\mu^2(a-b)^2)$ in dimensional regularization. We use the integrand.
- *Group I, Row V.* Parts 2a and 2b together. The $\log(\vee/\Lambda)$ part of 2b has the same two kernels as 2a, and the $N_c U^{af}(x)$ terms cancel.
- *Group II, Row IV.* The bracket $\frac{1}{\bar\epsilon} - \log(4\pi^2\mu^2(w-w')^2)$ is written as the integrand $\frac{1}{\pi}[K(w,w',z) - \frac{1}{2}K(w,w,z) - \frac{1}{2}K(w',w',z)]$, which has the same integral.
- *Group III, Row III, part 2.* The overall sign is reversed (fix F5).

After the colour algebra, and collecting equal terms within each row:

$$\frac{d^3N^{2\rho}_{\rm gen}}{d^3k}\Big|_{\log(\vee/\Lambda)}\Big/\mathcal{N}_0 L_\Lambda = \int e^{-ik(w'-w)}\Bigg[$$

$$\underline{\text{Group I, Row II (fix F6)}}$$
$$- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$$

$$
\begin{aligned}
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\; U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

Group I, Row V

$$
\begin{aligned}
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

Group II, Row I

$$
\begin{aligned}
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)
\end{aligned}
$$

Group II, Row IV

$$
\begin{aligned}
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&\underline{\text{Group III, Row III, part 2 (fix F5)}}\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&- 4\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{2.1}
$$

2.3 (c) The two-ρ lists of `4rho.tex`

These are your lists "Two ρ contribution" of `4rho.tex`, with fixes F1 (drop the $-8N_c$ term) and F2 (the repeated index in $dN^{(1)}$, term 2). Collected:

$$
\frac{d^3N_{4\rho}^{2\rho}}{d^3k}\Big|_{\log(\vee/\Lambda)}\Big/\mathcal{N}_0L_\Lambda=\int e^{-ik(w'-w)}\Big[
$$

$$
\begin{aligned}
&\underline{\texttt{4rho.tex}\text{ (fixes F1, F2)}}\\
&+ 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- 3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&- 3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)\\
&+ 3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&- 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\ f^{abc}f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\ f^{abc}f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ U^{ab}(x')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^l(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^e(x)\,\rho^c(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{deh}\,U^{ad}(z)\,U^{eb}(w)\,U^{lh}(x')\,\rho^l(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{deh}\,U^{ad}(z)\,U^{be}(w')\,U^{lh}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x')\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\ f^{abc}f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x')\Big]
\end{aligned}
\tag{2.2}
$$

The complete two-ρ content of the four-ρ terms is this list plus the two-ρ terms of the remainders d^3N_a, d^3N_b. The remainders are part of set (a).

3 JIMWLK⊗LO

Operator form. With $\delta Y = L_\Lambda$, the JIMWLK evolution of the LO cross section is

$$
\begin{aligned}
\mathcal{N}\,\mathcal{J} =&\ -\mathcal{N}\int e^{-ik\cdot(w'-w)}\int_{x,x'}\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\ \tilde{H}\Big\{\rho^{b'}(x')\big[U(x')-U(w')\big]^{ab'}\big[U(x)-U(w)\big]^{ab}\rho^b(x)\Big\},\\
\tilde{H} =&\int_{u,v,z}K(u,v,z)\Big\{J_R^h(u)J_R^h(v)+J_L^m(u)J_L^m(v)-U^{mh}(z)\big[J_R^h(u)J_L^m(v)+J_L^m(u)J_R^h(v)\big]\Big\},
\end{aligned}
\tag{3.1}
$$

with $K(u,v,z)=\mathcal{K}(u-z)\cdot\mathcal{K}(v-z)$ and the definitions below.

- $J_R^h(u)\,U(p)=\delta^{(2)}(u-p)\,U(p)T^h$, $J_L^m(u)=U^{mg}(u)J_R^g(u)$, $(T^h)_{bc}=-if^{hbc}$.
- This is H_{JIMWLK} of your Eq. (HLO), with $d/dY=-H$.
- $\tilde{H}$ differentiates the Wilson lines only. So u and v are each one of the points $x,\ x',\ w,\ w'$ where the LO cross section has a Wilson line: the soft gluon at z connects two of them.
- This gives the kernel structures $K(x,x,z)$, $K(x',x',z)$, $K(x,x',z)$ (soft gluon between the charges), $K(w,w,z)$, $K(w',w',z)$, $K(w,w',z)$ (between the measured gluons), and the mixed $K(x,w',z)$, $K(x',w,z)$.

Written out. Carrying out the derivatives and the colour algebra:

$$
\begin{aligned}
\mathcal{J} =&\int e^{-ik(w'-w)}\Bigg[\\
&+\underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}}_{S_1}\Big[\\
&\qquad+2\,N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\qquad-2\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\qquad+2\,f^{abc}f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)\\
&\qquad-2\,f^{abc}f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}}_{S_2}\Big[\\
&\quad + 2\,N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)}_{S_3}\Big[\\
&\quad + 2\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}}_{S_4}\Big[\\
&\quad - 2\,N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}}_{S_5}\Big[\\
&\quad - 2\,N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)}_{S_6}\Big[\\
&\quad + 2\,N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)}_{S_8}\Big[\\
&\quad - 2\,N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\Big]\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)}_{S_9}\Big[\\
&\quad - 2\,N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&\quad - 2\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&\quad + 2\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)\Big]\Big]
\end{aligned}
\tag{3.2}
$$

The terms of the structures that do not appear here cancel within the three sets.

4 The sum, structure by structure

4.1 How the comparison is done

1. Every term of (a), (b), (c) is written with its kernel in one of the 13 structures S_n of Table 1.
2. Within a structure, the colour structures are not all independent: Jacobi-type identities relate them (App. A). Each colour structure is written in a basis of independent ones, with coefficients that are exact polynomials in N_c (fixed from SU(3) and SU(4)).
3. For each basis colour structure, the coefficients from (a), (b), (c) are added and compared with $\mathcal{J}$.

	kernel structure	in $\mathcal{J}$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}$	✓
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}$	✓
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	✓
S_4	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}$	✓
S_5	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}$	✓
S_6	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$	✓
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$	–
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)$	✓
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)$	✓
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)$	–
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	–
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$	–
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)$	–

Table 1: The kernel structures of the $\log(\vee/\Lambda)$ two-ρ terms.

4.2 Worked example 1: the soft gluon between the two charges

The kernel is $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$ (structure S_3). Every term below multiplies it.

From the three-ρ equation (all rows).

$$
\begin{aligned}
&- f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&- f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&- f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+ f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&- f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^{c}(x')\,\rho^{e}(x)\\
&+ f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

Genuine two-ρ terms.

$$
\begin{aligned}
&+ \tfrac{1}{2}\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \tfrac{1}{2}\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+ \tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

Two-ρ lists of `4rho.tex`.

$$
\begin{aligned}
&+ 2\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^{c}(x')\,\rho^{e}(x)\\
&+ \tfrac{3}{2}\,N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&- f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+ \tfrac{1}{2}\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&- f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&+ f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^{c}(x')\,\rho^{e}(x)\\
&- f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

Sum of the three sets, in independent colour structures, and JIMWLK⊗LO.

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK⊗LO
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	$\frac{N_c}{2}$	$\frac{3N_c}{2}$	$2N_c$	$2N_c$
$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	0	0	2	2	2
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$	-1	0	-1	-2	-2
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	-1	0	-1	-2	-2
$f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$	-1	$\frac{1}{2}$	$\frac{1}{2}$	0	0
$f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)$	1	0	-1	0	0
$f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)$	-1	0	1	0	0
$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	0
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	0

4.3 Worked example 2: a structure that is not in JIMWLK

The kernel is $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$ (structure S_7). Every term below multiplies it.

From the three-ρ equation (all rows).

$$
\begin{aligned}
&+ N_c\; U^{ab}(x')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\; U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)\\
&-\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)\\
&+\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

Genuine two-ρ terms.

$$
\begin{aligned}
&- N_c\; U^{ab}(x')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\; U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)\\
&+\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)\\
&-\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

Two-ρ lists of `4rho.tex`. No terms with this kernel.

Sum of the three sets, in independent colour structures, and JIMWLK⊗LO.

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK⊗LO
$U^{ab}(x')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$	N_c	$-N_c$	0	0	0
$U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	N_c	0	0	0
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$	-1	1	0	0	0
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$	1	-1	0	0	0

4.4 All structures

Columns: the coefficient from (a) the three-ρ equation, (b) the genuine terms, (c) `4rho.tex`, their sum, and the coefficient in $\mathcal{J}$. All are in units of $\mathcal{N}$, multiplying [kernel]×[colour structure].

Structure S_1: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	0	$2N_c$	$2N_c$	$2N_c$ ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	0	$-2N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$	1	0	1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$	-1	0	-1	-2	-2 ✓

Structure S_2: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	0	$2N_c$	$2N_c$	$2N_c$ ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	0	$-2N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$	1	0	1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$	-1	0	-1	-2	-2 ✓

Structure S_3: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	$\frac{N_c}{2}$	$\frac{3N_c}{2}$	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	0	0	2	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$	-1	0	-1	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	-1	0	-1	-2	-2 ✓
$f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$	-1	$\frac{1}{2}$	$\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)$	1	0	-1	0	0 ✓
$f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)$	-1	0	1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	0 ✓

Structure S_4: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$-2N_c$	0	$-2N_c$	$-2N_c$ ✓
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$2N_c$	0	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$	0	-2	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$	0	2	0	2	2 ✓

Structure S_5: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(w'-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$-2N_c$	0	$-2N_c$	$-2N_c$ ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	$2N_c$	0	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$	0	-2	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$	0	2	0	2	2 ✓

Structure S_6: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$2N_c$	0	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	0	2	0	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$	0	-2	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$	0	-2	0	-2	-2 ✓

Structure S_7: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$

JIMWLK has no term with this kernel; the terms of the three sets cancel.

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$	N_c	$-N_c$	0	0	0 ✓
$U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	N_c	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$	-1	1	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$	1	-1	0	0	0 ✓

Structure S_8: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	$-N_c$	0	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	-2	0	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$	2	0	0	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	1	1	0	2	2 ✓
$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)$	-1	1	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$	-1	1	0	0	0 ✓

Structure S_9: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	$-N_c$	0	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	-2	0	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$	1	1	0	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$	2	0	0	2	2 ✓
$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')$	1	-1	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')$	1	-1	0	0	0 ✓

Structure S_{10}: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)$

JIMWLK has no term with this kernel; the terms of the three sets cancel.

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x)\,U^{ca}(w')\,\rho^c(x')\,\rho^b(x)$	N_c	$-N_c$	0	0	0 ✓
$U^{ab}(w)\,U^{ca}(w')\,\rho^c(x')\,\rho^b(x)$	$-N_c$	N_c	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{bd}(z)\,U^{ce}(w')\,\rho^h(x')\,\rho^l(x)$	-1	1	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{bd}(z)\,U^{ce}(w')\,\rho^h(x')\,\rho^l(x)$	1	-1	0	0	0 ✓

Structure S_{11}: $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$

Every colour structure has coefficient 0 in each of the three sets and in JIMWLK.

Structure S_{12}: $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$

Every colour structure has coefficient 0 in each of the three sets and in JIMWLK.

Structure S_{13}: $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)$

Every colour structure has coefficient 0 in each of the three sets and in JIMWLK.

Part II
BFKL: the coefficient of $\log\vee$

5 What has to be added

Write every row as

$$\frac{d^3N_{\rm row}}{d^3k} = \mathcal{N}_0\big[A_{\rm row}\,L_\Lambda + B_{\rm row}\,L_k\big] + (\text{no large log}), \qquad P_{\rm row} = A_{\rm row} + B_{\rm row}.$$

- $A_{\rm row}$ are the $\log(\vee/\Lambda)$ terms of Part I.
- $B_{\rm row}$ are the $\log(\vee/k^+)$ terms of the same row. In your notes they come from functions such as $\log(\Lambda/k^+)$, $\log\frac{\vee}{\Lambda+k^+}$ and $\log(\frac{\vee}{k^+}-1)$, or hidden in the plus-prescription integrals.
- $P_{\rm row}$ is the coefficient of $\log\vee$: the integrand of the row at the $\vee$ end of its p^+ integral.

For the two-ρ terms:

- The two-ρ terms from the four-ρ terms (the remainders d^3N_a, d^3N_b and `4rho.tex`) have $P=A$: their p^+ integral is $\int_\Lambda^\vee dp^+/p^+$ with a p^+-independent factor.
- Group I Row V, $\mathcal{D}$ and the genuine Group I terms have $P=0$: their p^+ integral runs from Λ to $k^+-\Lambda$, so it has no $\vee$.
- Group I Row II has only the $\mathfrak{F}_\alpha$ part, with the sign reversed (fix F4: this part multiplies $-\log(\vee/k^+)$).
- Every other row has its $\vee$-end integrand. Fix P1 (Group I Row III) and fix F5 (Group III Row III, part 2) apply.

The symmetrized three-ρ parts of P cancel (the $\log(\vee/k^+)$ note), so only two-ρ terms remain:

$$P/\mathcal{N}_0 = \int e^{-ik(w'-w)}\Bigg[$$

Group I, Row II (fix F4: the $\mathfrak{F}_\alpha$ part, with the sign reversed)

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')
\end{aligned}
$$

Group I, Row III (with fix P1)

$$
\begin{aligned}
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&- \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

Group I, Row IV

$$
- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)
$$

$$
\begin{aligned}
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\, U^{ab}(x')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\, f^{abc}\, f^{cde}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^l(x')\, \rho^e(x) \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\, U^{ab}(w')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, f^{abc}\, f^{dce}\, U^{hl}(x)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^e(x')\, \rho^l(x) \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, U^{ab}(x)\, U^{ac}(x')\, \rho^c(x')\, \rho^b(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, f^{abc}\, f^{dce}\, U^{hl}(w)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^e(x')\, \rho^l(x) \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, U^{ab}(w)\, U^{ac}(x')\, \rho^c(x')\, \rho^b(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ea}(w)\, U^{db}(z)\, U^{hl}(x')\, \rho^l(x')\, \rho^c(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{da}(w')\, U^{eb}(z)\, U^{hl}(x)\, \rho^c(x')\, \rho^l(x) \\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, U^{ab}(x)\, U^{ac}(x')\, \rho^c(x')\, \rho^b(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{le}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{md}(x')\, \rho^h(x')\, \rho^c(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ld}(w)\, U^{la}(w')\, U^{mb}(z)\, U^{me}(x)\, \rho^c(x')\, \rho^h(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{bd}(x')\, U^{cl}(x)\, \rho^h(x')\, \rho^l(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{be}(x)\, U^{cl}(x')\, \rho^l(x')\, \rho^h(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^m(x)\, \rho^e(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(x)\, \rho^m(x)\, \rho^e(x') \\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, U^{ab}(x)\, U^{ac}(x')\, \rho^b(x)\, \rho^c(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^m(x)\, \rho^e(x') \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{dce}\, U^{hd}(w)\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(x)\, \rho^m(x)\, \rho^e(x') \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{bl}(x)\, U^{cd}(x')\, \rho^l(x)\, \rho^h(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{bl}(x)\, U^{ce}(x')\, \rho^l(x)\, \rho^h(x') \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\, f^{abc}\, f^{cde}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^l(x')\, \rho^e(x) \\
&- N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\, U^{ab}(x')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\, f^{abc}\, f^{cde}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^l(x')\, \rho^e(x) \\
&+ N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\, U^{ab}(w')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ea}(w)\, U^{db}(z)\, U^{hl}(x')\, \rho^l(x')\, \rho^c(x) \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^m(x)\, \rho^e(x') \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{le}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{md}(x')\, \rho^h(x')\, \rho^c(x) \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^m(x)\, \rho^e(x') \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\, f^{abc}\, f^{dce}\, U^{hl}(x)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^e(x')\, \rho^l(x) \\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\, U^{ab}(x)\, U^{ac}(x')\, \rho^c(x')\, \rho^b(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\, f^{abc}\, f^{dce}\, U^{hl}(w)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^e(x')\, \rho^l(x) \\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\, U^{ab}(w)\, U^{ac}(x')\, \rho^c(x')\, \rho^b(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{da}(w')\, U^{eb}(z)\, U^{hl}(x)\, \rho^c(x')\, \rho^l(x) \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{dce}\, U^{hd}(x)\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(x')\, \rho^m(x')\, \rho^e(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{deh}\, U^{ld}(w)\, U^{la}(w')\, U^{mb}(z)\, U^{me}(x)\, \rho^c(x')\, \rho^h(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{dce}\, U^{hd}(w)\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(x')\, \rho^m(x')\, \rho^e(x)
\end{aligned}
$$

Group II, Row I

$$
\begin{aligned}
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\, U^{ab}(w')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{eb}(z)\, U^{cd}(x)\, \rho^l(x')\, \rho^h(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\, f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{lm}(x')\, U^{hd}(x)\, U^{la}(w)\, \rho^m(x')\, \rho^e(x) \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\, U^{ab}(x')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, U^{ab}(w')\, U^{ac}(w)\, \rho^c(x)\, \rho^b(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{bl}(z)\, U^{hm}(x)\, U^{ld}(x')\, \rho^m(x)\, \rho^e(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, f^{abc}\, f^{deh}\, U^{db}(z)\, U^{ce}(x')\, U^{al}(w)\, \rho^l(x)\, \rho^h(x') \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\, U^{ab}(x)\, U^{ac}(x')\, \rho^b(x)\, \rho^c(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\, f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^b(x)\, \rho^e(x')
\end{aligned}
$$

$$
\begin{aligned}
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{lb}(z)\, U^{cd}(x')\, \rho^{l}(x)\, \rho^{h}(x') \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{ld}(x')\, U^{hm}(x)\, U^{la}(w)\, \rho^{m}(x)\, \rho^{e}(x') \\
&+ N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{ac}(x')\, \rho^{b}(x)\, \rho^{c}(x') \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{e}(x')\, \rho^{b}(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{bl}(z)\, U^{hd}(x')\, U^{lm}(x)\, \rho^{e}(x')\, \rho^{m}(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{lb}(z)\, U^{ce}(x')\, U^{ad}(w)\, \rho^{h}(x')\, \rho^{l}(x) \\
&- N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; U^{ab}(w')\, U^{ac}(w)\, \rho^{b}(x')\, \rho^{c}(x) \\
&-\; \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{deh}\, U^{al}(w')\, U^{eb}(z)\, U^{cd}(x)\, \rho^{l}(x')\, \rho^{h}(x) \\
&-\; \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{lm}(x')\, U^{hd}(x)\, U^{la}(w)\, \rho^{m}(x')\, \rho^{e}(x) \\
&- N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&-\; \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{bde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^{e}(x')\, \rho^{c}(x) \\
&-\; \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{lb}(z)\, U^{cd}(x')\, \rho^{l}(x)\, \rho^{h}(x') \\
&-\; \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{ld}(x')\, U^{hm}(x)\, U^{la}(w)\, \rho^{m}(x)\, \rho^{e}(x') \\
&+ N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; U^{ab}(w')\, U^{ac}(w)\, \rho^{c}(x)\, \rho^{b}(x') \\
&+\; \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{bl}(z)\, U^{hm}(x)\, U^{ld}(x')\, \rho^{m}(x)\, \rho^{e}(x') \\
&+\; \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{deh}\, U^{db}(z)\, U^{ce}(x')\, U^{al}(w)\, \rho^{l}(x)\, \rho^{h}(x') \\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; U^{ab}(x)\, U^{ac}(x')\, \rho^{b}(x)\, \rho^{c}(x') \\
&-\; \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{bde}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{c}(x')\, \rho^{e}(x) \\
&+\; \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{bl}(z)\, U^{hd}(x)\, U^{lm}(x')\, \rho^{e}(x)\, \rho^{m}(x') \\
&+\; \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{lb}(z)\, U^{ce}(x)\, U^{ad}(w)\, \rho^{h}(x)\, \rho^{l}(x') \\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x)
\end{aligned}
$$

Group III, Row I

$$
\begin{aligned}
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{hl}(x')\, U^{md}(x)\, U^{mb}(z)\, U^{ha}(w)\, \rho^{l}(x')\, \rho^{e}(x) \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{hd}(x)\, U^{lm}(w')\, U^{la}(w)\, \rho^{m}(x')\, \rho^{e}(x) \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; U^{ab}(w')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{hl}(x)\, U^{md}(x')\, U^{mb}(z)\, U^{ha}(w')\, \rho^{e}(x')\, \rho^{l}(x) \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{bh}(z)\, U^{hd}(x')\, U^{lm}(w)\, U^{la}(w')\, \rho^{e}(x')\, \rho^{m}(x) \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; U^{ab}(w)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{db}(z)\, U^{ea}(w)\, U^{hl}(x')\, \rho^{l}(x')\, \rho^{c}(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{eb}(z)\, U^{da}(w')\, U^{hl}(x)\, \rho^{c}(x')\, \rho^{l}(x) \\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{bl}(z)\, U^{ld}(x')\, U^{me}(w')\, U^{ma}(w)\, \rho^{h}(x')\, \rho^{c}(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{bl}(z)\, U^{le}(x)\, U^{md}(w)\, U^{ma}(w')\, \rho^{c}(x')\, \rho^{h}(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{bd}(x')\, U^{ae}(w')\, U^{cl}(x)\, \rho^{h}(x')\, \rho^{l}(x) \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{be}(x)\, U^{ad}(w)\, U^{cl}(x')\, \rho^{l}(x')\, \rho^{h}(x) \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{lm}(x)\, U^{lb}(z)\, U^{ha}(w)\, \rho^{m}(x)\, \rho^{e}(x') \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{lm}(x)\, U^{lb}(z)\, U^{ha}(w')\, \rho^{m}(x)\, \rho^{e}(x') \\
&- N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{ac}(x')\, \rho^{b}(x)\, \rho^{c}(x') \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{hl}(x)\, U^{md}(w')\, U^{ma}(w)\, \rho^{l}(x)\, \rho^{e}(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{bh}(z)\, U^{hl}(x)\, U^{md}(w)\, U^{ma}(w')\, \rho^{l}(x)\, \rho^{e}(x') \\
&+ \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{bl}(x)\, U^{ae}(w')\, U^{cd}(x')\, \rho^{l}(x)\, \rho^{h}(x') \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{bl}(x)\, U^{ad}(w)\, U^{ce}(x')\, \rho^{l}(x)\, \rho^{h}(x') \\
&+\; \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{hl}(x')\, U^{md}(x)\, U^{mb}(z)\, U^{ha}(w)\, \rho^{l}(x')\, \rho^{e}(x) \\
&+ N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&-\; \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{hd}(x)\, U^{lm}(w')\, U^{la}(w)\, \rho^{m}(x')\, \rho^{e}(x)
\end{aligned}
$$

$$
\begin{aligned}
&- N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ea}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ha}(w)\,\rho^m(x)\,\rho^e(x')\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{ma}(w)\,\rho^h(x')\,\rho^c(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{ma}(w)\,\rho^l(x)\,\rho^e(x')\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ha}(w')\,\rho^e(x')\,\rho^l(x)\\
&+ N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(x')\,U^{lm}(w)\,U^{la}(w')\,\rho^e(x')\,\rho^m(x)\\
&- N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{da}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{lm}(x')\,U^{lb}(z)\,U^{ha}(w')\,\rho^m(x')\,\rho^e(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(x')\,U^{md}(w)\,U^{ma}(w')\,\rho^l(x')\,\rho^e(x)
\end{aligned}
$$

Group III, Row II

$$
\begin{aligned}
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\rho^m(x')\,\rho^e(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{la}(w')\,\rho^m(x)\,\rho^e(x')\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+ \tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{be}(z)\,U^{da}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\rho^l(x)\,\rho^e(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w')\,\rho^e(x')\,\rho^l(x)\\
&- N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,\rho^e(x')\,\rho^b(x)\\
&+ N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\rho^m(x')\,\rho^e(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\rho^l(x)\,\rho^e(x')\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{be}(z)\,U^{da}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&- 2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{la}(w')\,\rho^m(x)\,\rho^e(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{md}(x)\,U^{bh}(z)\,U^{ma}(w')\,\rho^{e}(x)\,\rho^{l}(x')\\
&- 2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{c}(x')\,\rho^{e}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{bc}(x)\,\rho^{c}(x)\,\rho^{a}(x')
\end{aligned}
$$

Group III, Row III, part 1

$$
\begin{aligned}
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(x')\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{cd}(x)\,U^{bl}(x')\,\rho^{h}(x)\,\rho^{l}(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ae}(x)\,U^{cd}(w)\,\rho^{h}(x)\,\rho^{l}(x')\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)
\end{aligned}
$$

Group III, Row V

$$
\begin{aligned}
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{be}(x)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(w')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{bl}(x)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{la}(z)\,U^{ce}(w')\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(w')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{cl}(w)\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{bl}(x')\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{la}(z)\,U^{cd}(w)\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{cd}(w)\,U^{bl}(x')\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

Group III, Row VI

$$
\begin{aligned}
&+ 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+ 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- 2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\\
&- 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')
\end{aligned}
$$

$$+\,2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$$
$$-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')$$
$$-\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$-\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$$
$$-\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(z)\,U^{cd}(x)\,U^{bl}(x')\,\rho^h(x)\,\rho^l(x')$$
$$+\,2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)$$
$$+\,2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$

Four-ρ remainders d^3N_a, d^3N_b $(P=A)$

$$-\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$$
$$+\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$$
$$+\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$-\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$$
$$-\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$$
$$+\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$-\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$$
$$+\,\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)$$
$$-\,\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$$
$$-\,\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)$$
$$+\,\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$$
$$-\,\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')$$
$$+\,\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hd}(x')\,\rho^a(x)\,\rho^e(x')$$
$$-\,\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{hd}(x')\,\rho^a(x)\,\rho^e(x')$$
$$+\,\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{hd}(w)\,\rho^a(x)\,\rho^e(x')$$
$$-\,\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')$$
$$-\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$+\,\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)$$

Genuine: Group II, Row I

$$+\,\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$$
$$-\,\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$$
$$-\,\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$$
$$+\,\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$$
$$+\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')$$
$$+\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$$
$$-\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x')$$
$$-\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$$
$$-\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^l(x)\,\rho^h(x')$$
$$-\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$$
$$+\,2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$$
$$+\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$$
$$-\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$$
$$-\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$$
$$+\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$$
$$+\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$$
$$-\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$$
$$-\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$$

$$
\begin{aligned}
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&\underline{\text{Genuine: Group II, Row IV}}\\
&+ 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')\\
&- 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')\\
&- 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&+ 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')\\
&\underline{\text{Genuine: Group III, Row III, part 2 (fix F5)}}\\
&+\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')\\
&+\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&+ 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+ 2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x)\,U^{cd}(w)\,\rho^l(x)\,\rho^h(x')\\
&-\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&-\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(w')\,\rho^h(x)\,\rho^l(x')\\
&-\ \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&- 4\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&+ N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&-\ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&- N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+ N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&- N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{5.1}
$$

The genuine Group II Row IV term is the $\vee$ end of the same p^+ integral whose Λ end gives the genuine term of Part I:

- at the Λ end, the soft gluon at z is emitted by the measured gluon ($K(w,w',z)$, JIMWLK);
- at the $\vee$ end, the gluon at z is the parent and emits the measured gluon at w,w' ($\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,K(x,x',z)$, BFKL).

6 BFKL⊗LO

Operator form. Let $O(x',x)=\rho^{b'}(x')\,M^{b'b}(x',x)\,\rho^b(x)$ with $M^{b'b}(p',p)=\big[U(w')-U(p')\big]^{ab'}\big[U(w)-U(p)\big]^{ab}$. The LO cross section is $\propto\int\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\langle O(x',x)\rangle$. A gluon harder than the measured one is added to the projectile: its charge $j^a(z)$ adds to ρ^a, and the valence charges get the virtual correction. Then

$$
\begin{aligned}
\mathcal{B}=2\int_{x,x',u,v,z} e^{-ik\cdot(w'-w)}\,K(u,v,z)\,\Big\{
&-\tfrac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\Big[\rho^h(u)\rho^h(v)\,O(x',x)-2\rho^h(u)\,O(x',x)\,\rho^h(v)+O(x',x)\,\rho^h(u)\rho^h(v)\Big]\\
&+\mathcal{K}(z-w')\cdot\mathcal{K}(z-w)\;\rho^{e'}(u)\rho^e(v)\,(T^{b'}T^b)_{e'e}\,M^{b'b}(z,z)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\;\rho^{e'}(u)\rho^{b'}(x')\rho^e(v)\,(T^b)_{e'e}\,M^{b'b}(x',z)\\
&+\mathcal{K}(z-w')\cdot\mathcal{K}(x-w)\;\rho^{e'}(u)\rho^b(x)\rho^e(v)\,(T^{b'})_{e'e}\,M^{b'b}(z,x)\Big\}.
\end{aligned}
\tag{6.1}
$$

- *First line:* the virtual correction of the valence charges (dipole kernel). Here u, v are charges.
- *Second line:* both measured gluons are emitted by the gluon at z.
- *Last two lines:* one measured gluon is emitted by a valence charge and the other by the gluon at z.

In the colour-singlet projection this is Eq. (LOBFKL) of your Appendix E. After the charges are reordered, its three- and four-ρ parts vanish. Written out:

$$
\begin{aligned}
\mathcal{B} = \int e^{-ik(w'-w)} \Bigg[\\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}}_{S_1'} \Big[\\
&\qquad - N_c\, U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&\qquad + N_c\, U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&\qquad + N_c\, U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&\qquad - N_c\, U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \Big] \\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}}_{S_2'} \Big[\\
&\qquad - N_c\, U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&\qquad + N_c\, U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&\qquad + N_c\, U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&\qquad - N_c\, U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \Big] \\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)}_{S_3'} \Big[\\
&\qquad + 2\, f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
&\qquad - 2\, f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) \\
&\qquad - 2\, f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) \\
&\qquad + 2\, f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) \Big] \\
&+ \underbrace{\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)}_{S_5'} \Big[\\
&\qquad + 2\, N_c\, \rho^a(x)\,\rho^a(x') \\
&\qquad - \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') \\
&\qquad - \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) \\
&\qquad + \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x') \\
&\qquad + \ f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x) \\
&\qquad + \ f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') \\
&\qquad + \ f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) \Big] \\
&+ \underbrace{\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}}_{S_6'} \Big[\\
&\qquad - N_c\, U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&\qquad + N_c\, U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \\
&\qquad + N_c\, U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&\qquad - N_c\, U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) \Big] \\
&+ \underbrace{\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)}_{S_7'} \Big[\\
&\qquad + 2\, f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&\qquad - 2\, f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x') \\
&\qquad - 2\, f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&\qquad + 2\, f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x') \Big] \\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}}_{S_8'} \Big[\\
&\qquad - N_c\, U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')
\end{aligned}
$$

$$
\begin{aligned}
&+ N_c\, U^{ab}(w')\, U^{ac}(x)\, \rho^c(x)\, \rho^b(x') \\
&+ N_c\, U^{ab}(z)\, U^{ac}(w)\, \rho^c(x)\, \rho^b(x') \\
&- N_c\, U^{ab}(z)\, U^{ac}(x)\, \rho^c(x)\, \rho^b(x') \Big] \\
&+ \underbrace{\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)}_{S'_9} \Big[\\
&+ 2\, f^{abc} f^{cde}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x) \\
&- 2\, f^{abc} f^{cde}\, U^{ha}(w')\, U^{hd}(x)\, \rho^b(x')\, \rho^e(x) \\
&- 2\, f^{abc} f^{cde}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x) \\
&+ 2\, f^{abc} f^{cde}\, U^{ha}(z)\, U^{hd}(x)\, \rho^b(x')\, \rho^e(x) \Big] \Big]
\end{aligned}
\tag{6.2}
$$

7 The sum, structure by structure

Same method as Sec. 4, now for P against $\mathcal{B}$. The columns are the $\vee$-end terms of the three-ρ rows, of the genuine rows, and of `4rho.tex` ($P = A$).

	kernel structure	in $\mathcal{B}$
S'_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}$	✓
S'_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}$	✓
S'_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	✓
S'_4	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	–
S'_5	$\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	✓
S'_6	$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}$	✓
S'_7	$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	✓
S'_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}$	✓
S'_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	✓

Table 2: The kernel structures of the log $\vee$ two-ρ terms.

7.1 Worked example 1: both measured gluons from the gluon at z

The kernel is $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$ (structure S'_5). Every term below multiplies it.

Three-ρ rows ($\vee$ end). No terms with this kernel.

Genuine rows ($\vee$ end).

$$
\begin{aligned}
&- f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x)\, \rho^e(x') \\
&- f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x) \\
&+ f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(z)\, \rho^b(x)\, \rho^e(x') \\
&+ f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(z)\, \rho^b(x')\, \rho^e(x) \\
&+ f^{abc} f^{dce}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x)\, \rho^e(x') \\
&+ f^{abc} f^{dce}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x) \\
&+ 2\, N_c\, \rho^a(x)\, \rho^a(x')
\end{aligned}
$$

Two-ρ lists of `4rho.tex` ($P = A$). No terms with this kernel.

Sum of the three sets, in independent colour structures, and BFKL⊗LO.

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL⊗LO
$\rho^a(x)\,\rho^a(x')$	0	$2N_c$	0	$2N_c$	$2N_c$
$f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x)\, \rho^e(x')$	0	-1	0	-1	-1
$f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x)$	0	-1	0	-1	-1
$f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(z)\, \rho^b(x)\, \rho^e(x')$	0	1	0	1	1
$f^{abc} f^{dce}\, U^{ha}(w')\, U^{hd}(z)\, \rho^b(x')\, \rho^e(x)$	0	1	0	1	1
$f^{abc} f^{dce}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x)\, \rho^e(x')$	0	1	0	1	1
$f^{abc} f^{dce}\, U^{ha}(z)\, U^{hd}(w)\, \rho^b(x')\, \rho^e(x)$	0	1	0	1	1

This whole structure comes from the $\vee$ end of the genuine Group II, Row IV term (Sec. 5). It is the second line of Eq. (6.1). It has no counterpart in Part I: no $\log(\vee/\Lambda)$ term has the measured gluon emitted by the gluon at z.

7.2 Worked example 2: the soft gluon between the two charges

The kernel is $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$ (structure S_3'). Every term below multiplies it.

Three-ρ rows ($\vee$ end).

$$
\begin{aligned}
&+\ f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-\ f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&-\ f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-\ f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\ f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-2\,N_c\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

Genuine rows ($\vee$ end).

$$
\begin{aligned}
&+\tfrac{1}{2}\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\tfrac{1}{2}\,N_c\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&-\tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

Two-ρ lists of `4rho.tex` ($P=A$).

$$
\begin{aligned}
&+\tfrac{1}{2}\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-\ f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&-\ f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&+\ f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&-\ f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{3}{2}\,N_c\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\tfrac{1}{2}\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\ f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

Sum of the three sets, in independent colour structures, and BFKL⊗LO.

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL⊗LO
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	1	$\frac{1}{2}$	$\frac{1}{2}$	2	2
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	-1	0	-1	-2	-2
$f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	-1	0	-1	-2	-2
$f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	0	0	2	2	2
$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$	$-2N_c$	$\frac{N_c}{2}$	$\frac{3N_c}{2}$	0	0
$f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	-1	0	1	0	0
$f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	0
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	0
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0

7.3 All structures

Structure S_1': $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	0	0	$-N_c$	$-N_c$ ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	0	$2N_c$	N_c	N_c ✓
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	N_c	0	0	N_c	N_c ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	N_c	0	$-2N_c$	$-N_c$	$-N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	-1	0	1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0 ✓

Structure S_2': $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	0	0	$-N_c$	$-N_c$ ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	N_c	0	0	N_c	N_c ✓
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	0	$2N_c$	N_c	N_c ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	N_c	0	$-2N_c$	$-N_c$	$-N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$	1	0	-1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$	-1	0	1	0	0 ✓

Structure S_3': $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	1	$\frac{1}{2}$	$\frac{1}{2}$	2	2 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	-1	0	-1	-2	-2 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	-1	0	-1	-2	-2 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	0	0	2	2	2 ✓
$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$	$-2N_c$	$\frac{N_c}{2}$	$\frac{3N_c}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	-1	0	1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0 ✓

Structure S_4': $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$

Every colour structure has coefficient 0 in each of the three sets and in BFKL.

Structure S_5': $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$\rho^a(x)\,\rho^a(x')$	0	$2N_c$	0	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$	0	-1	0	-1	-1 ✓
$f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$	0	-1	0	-1	-1 ✓
$f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x')$	0	1	0	1	1 ✓
$f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x)$	0	1	0	1	1 ✓
$f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')$	0	1	0	1	1 ✓
$f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$	0	1	0	1	1 ✓

Structure S_6': $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	0	0	$-N_c$	$-N_c$ ✓
$U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)$	N_c	0	0	N_c	N_c ✓
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	N_c	0	0	N_c	N_c ✓
$U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	0	0	$-N_c$	$-N_c$ ✓

Structure S_7': $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$	1	1	0	2	2 ✓
$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')$	-1	-1	0	-2	-2 ✓
$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$	-2	0	0	-2	-2 ✓
$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')$	2	0	0	2	2 ✓
$U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	$-N_c$	N_c	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$	-1	1	0	0	0 ✓

Structure S_8': $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')$	$-N_c$	0	0	$-N_c$	$-N_c$ ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')$	N_c	0	0	N_c	N_c ✓
$U^{ab}(z)\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')$	N_c	0	0	N_c	N_c ✓
$U^{ab}(z)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')$	$-N_c$	0	0	$-N_c$	$-N_c$ ✓

Structure S_9': $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$

colour structure	3ρ eq.	genuine	`4rho.tex`	sum	BFKL
$f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$	1	1	0	2	2 ✓
$f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)$	-2	0	0	-2	-2 ✓
$f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)$	-1	-1	0	-2	-2 ✓
$f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)$	2	0	0	2	2 ✓
$U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	N_c	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	1	-1	0	0	0 ✓

8 Putting the two together

- Part I: $\sum_{\text{rows}} A_{\text{row}} = \mathcal{J}$, the coefficient of $L_\Lambda = \log(\vee/\Lambda)$.
- Part II: $\sum_{\text{rows}} P_{\text{row}} = \mathcal{B}$, the coefficient of $\log\vee$.
- Since $P = A + B$, the coefficient of $L_k = \log(\vee/k^+)$ is $\sum_{\text{rows}} B_{\text{row}} = \mathcal{B} - \mathcal{J}$.

This is Eq. (1.2):

$$\left.\frac{d^3N}{d^3k}\right|_{\text{LL}}^{2\rho} = \mathcal{N}_0\Big[\log\frac{\vee}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,(\mathcal{B}-\mathcal{J})\Big] = \mathcal{N}_0\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}\Big].$$

The target evolves (JIMWLK) over the rapidity interval below the measured gluon, $\log(k^+/\Lambda)$. The projectile evolves (BFKL) over the interval above it, $\log(\vee/k^+)$. Together these make $\delta Y = \log(\vee/\Lambda)$.

A Colour identities used in the basis reduction

In each kernel structure where the colour structures are linearly dependent, every dependent colour structure equals the following combination of basis structures. The identities are exact and were checked for SU(3) and SU(4).

A.1 Part I

Structure S_1.

$$\begin{aligned}
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x)\,U^{lm}(w')\,U^{le}(x)\,\rho^m(x')\,\rho^c(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x)\,U^{lm}(w')\,U^{lb}(x)\,\rho^m(x')\,\rho^c(x)\\
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x)\,U^{lm}(x')\,U^{le}(x)\,\rho^m(x')\,\rho^c(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x)\,U^{lm}(x')\,U^{lb}(x)\,\rho^m(x')\,\rho^c(x)\\
&f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)\\
&\quad= -f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) + f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)\\
&f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)\\
&\quad= -f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) - f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)
\end{aligned}$$

Structure S_2.

$$\begin{aligned}
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x')\,U^{le}(x')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x')\,U^{lb}(x')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)\\
&f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(x')\,U^{le}(x')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)\\
&\quad= f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x) - f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(x')\,U^{lb}(x')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)
\end{aligned}$$

Structure S_4.

$$
\begin{aligned}
&f^{abc}\, f^{deb}\, U^{hd}(z)\, U^{ha}(w)\, U^{lm}(w')\, U^{le}(w)\, \rho^{m}(x')\, \rho^{c}(x) \\
&\quad = f^{abc}\, f^{deh}\, U^{da}(z)\, U^{el}(w')\, U^{hb}(w)\, \rho^{l}(x')\, \rho^{c}(x) - f^{abc}\, f^{dea}\, U^{hd}(z)\, U^{he}(w)\, U^{lm}(w')\, U^{lb}(w)\, \rho^{m}(x')\, \rho^{c}(x) \\
&f^{abc}\, f^{deb}\, U^{hd}(z)\, U^{ha}(w)\, U^{lm}(x')\, U^{le}(w)\, \rho^{m}(x')\, \rho^{c}(x) \\
&\quad = f^{abc}\, f^{deh}\, U^{da}(z)\, U^{el}(x')\, U^{hb}(w)\, \rho^{l}(x')\, \rho^{c}(x) - f^{abc}\, f^{dea}\, U^{hd}(z)\, U^{he}(w)\, U^{lm}(x')\, U^{lb}(w)\, \rho^{m}(x')\, \rho^{c}(x)
\end{aligned}
$$

Structure S_5.

$$
\begin{aligned}
&f^{abc}\, f^{deb}\, U^{hd}(z)\, U^{ha}(w')\, U^{le}(w')\, U^{lm}(w)\, \rho^{c}(x')\, \rho^{m}(x) \\
&\quad = f^{abc}\, f^{deh}\, U^{da}(z)\, U^{el}(w)\, U^{hb}(w')\, \rho^{c}(x')\, \rho^{l}(x) - f^{abc}\, f^{dea}\, U^{hd}(z)\, U^{he}(w')\, U^{lb}(w')\, U^{lm}(w)\, \rho^{c}(x')\, \rho^{m}(x) \\
&f^{abc}\, f^{deb}\, U^{hd}(z)\, U^{ha}(w')\, U^{le}(w')\, U^{lm}(x)\, \rho^{c}(x')\, \rho^{m}(x) \\
&\quad = f^{abc}\, f^{deh}\, U^{da}(z)\, U^{el}(x)\, U^{hb}(w')\, \rho^{c}(x')\, \rho^{l}(x) - f^{abc}\, f^{dea}\, U^{hd}(z)\, U^{he}(w')\, U^{lb}(w')\, U^{lm}(x)\, \rho^{c}(x')\, \rho^{m}(x)
\end{aligned}
$$

A.2 Part II

Structure S_1'.

$$
\begin{aligned}
&f^{abc}\, f^{cde}\, U^{ah}(z)\, U^{dl}(w)\, U^{lm}(w')\, U^{hb}(x)\, \rho^{m}(x')\, \rho^{e}(x) \\
&\quad = -f^{abc}\, f^{cde}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^{l}(x')\, \rho^{e}(x) + f^{abc}\, f^{cde}\, U^{ah}(w)\, U^{bl}(x)\, U^{dl}(z)\, U^{mh}(w')\, \rho^{m}(x')\, \rho^{e}(x) \\
&f^{abc}\, f^{cde}\, U^{hd}(x)\, U^{hl}(w')\, U^{ma}(z)\, U^{mb}(x)\, \rho^{l}(x')\, \rho^{e}(x) \\
&\quad = -f^{abc}\, f^{deh}\, U^{al}(w')\, U^{ce}(x)\, U^{bd}(z)\, \rho^{l}(x')\, \rho^{h}(x) + f^{abc}\, f^{cde}\, U^{ah}(z)\, U^{bl}(x)\, U^{dh}(x)\, U^{ml}(w')\, \rho^{m}(x')\, \rho^{e}(x) \\
&f^{abc}\, f^{cde}\, U^{hd}(x)\, U^{hl}(x')\, U^{ma}(z)\, U^{mb}(x)\, \rho^{l}(x')\, \rho^{e}(x) \\
&\quad = -f^{abc}\, f^{deh}\, U^{al}(x')\, U^{ce}(x)\, U^{bd}(z)\, \rho^{l}(x')\, \rho^{h}(x) + f^{abc}\, f^{cde}\, U^{ah}(z)\, U^{bl}(x)\, U^{dh}(x)\, U^{ml}(x')\, \rho^{m}(x')\, \rho^{e}(x) \\
&f^{abc}\, f^{cde}\, U^{hb}(w)\, U^{hl}(x')\, U^{ma}(x)\, U^{md}(z)\, \rho^{l}(x')\, \rho^{e}(x) \\
&\quad = -f^{abc}\, f^{cde}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^{l}(x')\, \rho^{e}(x) - f^{abc}\, f^{cde}\, U^{hd}(w)\, U^{hl}(x')\, U^{ma}(z)\, U^{mb}(x)\, \rho^{l}(x')\, \rho^{e}(x)
\end{aligned}
$$

Structure S_2'.

$$
\begin{aligned}
&f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{hl}(w)\, U^{ma}(z)\, U^{mb}(x')\, \rho^{l}(x)\, \rho^{e}(x') \\
&\quad = f^{abc}\, f^{deh}\, U^{al}(w)\, U^{cd}(x')\, U^{be}(z)\, \rho^{h}(x')\, \rho^{l}(x) + f^{abc}\, f^{cde}\, U^{hb}(x')\, U^{hl}(w)\, U^{ma}(z)\, U^{md}(x')\, \rho^{l}(x)\, \rho^{e}(x') \\
&f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{hl}(x)\, U^{ma}(z)\, U^{mb}(x')\, \rho^{l}(x)\, \rho^{e}(x') \\
&\quad = f^{abc}\, f^{deh}\, U^{al}(x)\, U^{cd}(x')\, U^{be}(z)\, \rho^{h}(x')\, \rho^{l}(x) + f^{abc}\, f^{cde}\, U^{hb}(x')\, U^{hl}(x)\, U^{ma}(z)\, U^{md}(x')\, \rho^{l}(x)\, \rho^{e}(x')
\end{aligned}
$$

Structure S_4'.

$$
\begin{aligned}
&f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^{b}(x)\, \rho^{e}(x') \\
&\quad = f^{abc}\, f^{cde}\, U^{hb}(w')\, U^{ha}(w)\, \rho^{d}(x)\, \rho^{e}(x') - f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{b}(x)\, \rho^{e}(x') \\
&f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(x')\, \rho^{b}(x)\, \rho^{e}(x') \\
&\quad = f^{abc}\, f^{cde}\, U^{hb}(w')\, U^{ha}(x')\, \rho^{d}(x)\, \rho^{e}(x') - f^{abc}\, f^{dce}\, U^{ha}(w')\, U^{hd}(x')\, \rho^{b}(x)\, \rho^{e}(x') \\
&f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^{b}(x)\, \rho^{e}(x') \\
&\quad = f^{abc}\, f^{cde}\, U^{hb}(x')\, U^{ha}(w)\, \rho^{d}(x)\, \rho^{e}(x') - f^{abc}\, f^{dce}\, U^{ha}(x')\, U^{hd}(w)\, \rho^{b}(x)\, \rho^{e}(x')
\end{aligned}
$$

B Scripts

- `evo_build.py`: the three sets and the targets, by kernel structure and colour structure, and the basis reduction.
- `evo_note.py`: writes this file from `evo_static.tex`.

Set (a) is read from `ThreeRho_final.tex` (black and blue lines). Sets (b), (c), the P terms and the targets come from the scripts of `../two_rho_all`, `../two_rho` and `../bfkl`.